MODULE 13 · DOMAIN 3 - APPLICATION DEVELOPMENT

Embedding models and context length

Choosing the model that turns text into vectors, sized to your documents, your queries and your budget.

SECTION 3 of 6 EXAM WEIGHT 30% MODULE 13 of 40

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The embedding window is the envelope. The chunk is the letter. **Too small and the end is cut off. Far too big and you pay to post air.**

Hosted embedding models, September 2026

MODEL	WINDOW	DIMENSIONS	NOTES
bge-large-en	512 tokens	1,024	English. Normalised
gte-large-en	8,192 tokens	1,024	English. Not normalised
qwen3-embedding-0-6b	about 32,000	up to 1,024	100+ languages. Preview

ON THE EXAM

Endpoint names start with databricks-. All three can back a Knowledge Assistant's AI Search index.

Three things decide the choice

01

Source documents

Chunk length, language, domain.

02

Expected queries

Length, language, wording.

03

Optimisation strategy

Quality, or cost, latency and storage.

04

Choose

The model that fits all three, then evaluate.

ON THE EXAM

The guide names all three inputs. A scenario usually stresses one. That one decides the answer.

Too small, or far too big?

WINDOW SHORTER THAN CHUNKS

Silent truncation

- End of every long chunk ignored
- No error is raised
- Later content never matches
- A real quality failure

WINDOW FAR LONGER THAN CHUNKS

No extra benefit

- Short chunks do not use it
- Retrieval is not better for it
- Model may cost more to run
- Not a failure, just no gain

THE TIEBREAKER

The window must cover your longest chunk. Beyond that, window size is not a reason to choose.

What the queries tell you

- Queries are usually **short** and chunks **long**. Retrieval is asymmetric by nature.
- BGE's authors suggest a **query instruction** such as: Represent this sentence for searching relevant passages.
- The **language** of queries matters as much as the language of documents.
- If users use different words from the documents, test hybrid search and domain fit.
- Long queries, such as a pasted email, need a window that fits them too.

> ON THE EXAM
>
> Look at real queries before choosing. Their length and language rule models in or out.

One model for documents and queries

- Documents and queries must be embedded by **the same model**.
- Vectors from different models live in **different spaces**. Distances between them mean nothing.
- Changing the embedding model means **re-embedding every chunk**.
- AI Search managed embeddings use the chosen endpoint for both sides.
- A cheaper model for queries only is not an optimisation. It breaks retrieval.

> ON THE EXAM
>
> Mixing embedding models is like measuring one side in miles and the other in kilometres, then comparing the numbers.

Dimensions, and what they cost at scale

10 MILLION CHUNKS AT	STORAGE, 4 BYTES PER VALUE
1,024 dimensions	about 41 GB
768 dimensions	about 31 GB
256 dimensions	about 10 GB

ON THE EXAM

Fewer dimensions mean less storage and faster search, usually at some cost in quality. Evaluate before cutting.

What each strategy points to

IF THE PRIORITY IS	CHOOSE
Quality on long, structured chunks	A window at least as long as the chunks, evaluated best
Cost, latency and storage at huge scale	Fewer dimensions, a smaller model, if evaluation allows
Many languages	A multilingual model, for documents and queries
A specialised vocabulary	A model that evaluates well on your domain
Cosine similarity	Normalised vectors. Normalise GTE yourself

ON THE EXAM

AI Search measures L2 distance. With normalised vectors, ranking by L2 matches ranking by cosine similarity.

Harbourline chooses its embedding model

01
Documents
Clauses up to 900 tokens, English.

02
Queries
Short customer questions, English.

03
Strategy
Quality first, 2 million chunks.

04
Rules out
BGE: 512 is shorter than 900.

05
Chooses
GTE large, normalised, same for queries.

ON THE EXAM

The longest chunk ruled one model out. Evaluation, not window size, chose between the rest.

> A team chunks technical manuals into sections of up to 1,200 tokens. It is choosing between hosted English embedding models.
>
> Nothing in a section may be left out of its vector.

Which choice meets the requirement?

A BGE large, keeping the same sections of up to 1,200 tokens

B Any hosted model, since truncation does not affect retrieval

C The model with the most dimensions, whatever its window size

D A model with a window of at least 1,200 tokens, such as GTE large

◆ PAUSE AND DECIDE

Which choice meets the requirement?

A BGE large

512-token window. Over half of each long section is dropped.

B Any model

Truncated text can never be matched. It does affect retrieval.

C Most dimensions

Dimensions do not extend the window.

D Window of 1,200 or more

Fits the longest chunk, so nothing is cut off.

THE SIGNAL

Up to 1,200 tokens, and nothing left out. The window must be at least as long as the longest chunk.

> To save money, an engineer embeds 5 million document chunks with a high-quality model, but embeds user queries with a much cheaper model at query time.
>
> Retrieval results now look almost random.

What is the fix?

A Retrieve more results per query to widen the net

B Embed queries with the same model as the documents

C Normalise the query vectors before each search

D Add a reranker to the query to reorder results

◆ PAUSE AND DECIDE

What is the fix?

A More results

More random results are still random.

B Same model for queries

Only vectors from one model are comparable.

C Normalise queries

Changes vector length, not which space it lives in.

D Reranker

Reorders a candidate set that was chosen at random.

THE SIGNAL

Two different embedding models, and results that look random. Vectors from different models cannot be compared.

> A company's support knowledge base and incoming tickets are written in twelve languages. Its index uses an English-only embedding model.
>
> Retrieval works well for English questions and badly for most others.

What should the engineer do?

A. Switch to a larger English-only embedding model

B. Translate the documents into English, but leave queries in their original language

C. Use a multilingual embedding model, and re-embed the whole index

D. Increase the chunk size

◆ PAUSE AND DECIDE

What should the engineer do?

A **Larger English-only model**
Still cannot represent the other eleven languages well.

B **Translate documents only**
Queries in eleven languages still meet an English-only model.

C **Multilingual model, re-embed**
Handles both sides in every language. Re-embed for one space.

D **Bigger chunks**
Language, not size, is the problem.

THE SIGNAL

Twelve languages, and good only in English. Language fit is part of choosing an embedding model, for both documents and queries.

> A retailer's product index will hold 500 million short chunks of about 200 tokens. Storage cost and query latency are the top priorities.
>
> Evaluation shows 256-dimension embeddings retrieve almost as well as 1,024-dimension ones for this data.

Which choice best fits the optimisation strategy?

A. A window comfortably above 200 tokens, at 256 dimensions

B. The longest-window model available, at 1,024 dimensions

C. Chunks of 8,000 tokens, to use a long window fully instead

D. Keep 1,024 dimensions, and add a reranker for quality

◆ PAUSE AND DECIDE

Which choice best fits the optimisation strategy?

A Window above 200, 256 dimensions

Fits the chunks, and a quarter of the storage.

B Longest window, 1,024 dims

Pays for window and dimensions the data does not need.

C 8,000-token chunks

Blurs vectors, breaks the product-per-chunk design.

D 1,024 dims plus reranker

Adds storage and latency, both stated priorities.

THE SIGNAL

Storage and latency are the priority, and evaluation says 256 dimensions is almost as good. Follow the evidence and the strategy.

What to carry forward

- Three inputs: **documents, queries, optimisation strategy**.
- Window **at least as long as the longest chunk**. Bigger beyond that brings no gain.
- **Same model** for documents and queries. Change model, re-embed everything.
- BGE 512, GTE 8,192, Qwen3 about 32,000 and multilingual. Language fit matters.
- Fewer dimensions for scale, if evaluation allows. Normalise for cosine.

> ON THE EXAM
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> Next: module 14 augments prompts with key fields and intents, and shifts a model from baseline to desired behaviour.